Construction and test of a small Time Projection Chamber

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The T2K experiment in Japan require new developments for the Time Projection Chamber in the near detector, the Padova INFN section contributed in the development. We used a small reproduction a small reproduction what is been mounting there. The initial main goal of the project was the construction from semi raw components of a small TPC, whihc was succesfull. We starting from the prototype we divided our work in different phases, first a preparation for the internal walls, the ìn the elecrical cabling has been permorfmed giving current to the instrument. At this point we created a simulation of the entire TPC and compared the electric field with a set of data we collected in the anode plane. After this we closed the TPC creating the gas system necessary, setting the readout electronics for the instrumented anode and generating the trigger using two scintillators. Last we performed a data analysis to control the quality. At the end starting from a simple prototype we were able to develop the entire TPC system. Thus in this report we are reporting the development of a new detector are we are giving important technical notes for possible next users.

CONTENTS

I. TIME PROJECTION CHAMBER WORKING PRINCIPLE

The Time Projection Chamber is a nowadays common apparatus for the tracking of charged particles. The working principle is the creation of ionization electrons by the passage of a charged particle in a medium, and the drift of said electrons in a uniform electric field towards a segmented anode. As of today TPCs are well developed and widely used: they can have a gaseous or liquid medium, and are employed in a variety of fields, such as tracking units in accelerator experiments and studies in dark matter. Their most significant features are the electric field distribution in space, the gas composition and the readout electronics, which are going to be taken into account in this report.

1. Definition of useful quantities

A. Energy loss of a particle in a medium

Charged particles lose their energy in a medium following the Bethe-Block equation:

$$\left\langle \frac{dE}{dx} \right\rangle = -\frac{4\pi e^4}{m_e c^2}\frac{Z}{\beta^2} n_e \ln\left(\frac{2m_e c^2\beta^2\gamma^2 W_{\max}}{I^2}\right)$$

CITAZIONE This quantity is important regarding the energy loss of the particles we are going to study. The formula describes a statistical energy loss, so regions with higher deposition can be found, and it is possible to produce δ electrons, which are energetic enough to produce a ionization track themselves. The Bethe-Block function is also used for identifying the particle crossing the TPC volume.

B. Drift velocity

We define the drift velocity in the presence of an electric and magnetic field using the Langevin equation:

$$\vec{v}_D(E, B) = \frac{\mu}{1+\mu^2 B^2}\left(\vec{E} + \mu\vec{E}\times\vec{B} + \mu^2\vec{B}(\vec{E}\cdot\vec{B})\right)$$

where μ is the mobility that depends on the electric and magnetic field. In modern TPCs the additional magnetic field can be used for particle identification, and does not interfere with the electronics functioning SIAMO SICURI???. Additionally, a transverse diffusion is present, due to the diffusion in the perpendicular plane compared to the drift direction. This generate an intrinsic width d of the track, which is quantified by

$$d = v_T \cdot \sqrt{L}$$

with $v_T \approx 275 \frac{\mu m}{\sqrt{cm}}$ for the specific gas we are using and L the distance anode track. CHE SAREBBE L? INOLTRE SERVIREBBE UNA REFERENCE DA DOVE HAI PRESO QUESTA TRATTAZIONE.

2. Track reconstruction

Let us assume that an ionization electron is created in a certain point P. We observe the drift of this electron towards a segmented anode, and the signal it creates allows us to determine two coordinates of P. The third coordinate can be calculated knowing the drift velocity of electrons in the medium. A scintillator measures the impact time of the incoming particle and acts as a trigger, and from the time taken by the ionization electrons to reach the anode we can infer the third coordinate of the particle, and reconstruct the position of P in three dimensions. This can be made for every point belonging to the trajectory of the ionizing particle, and allows us to reconstruct it fully in three dimensions.
Figure 1 shows this process visually; coordinates y and z are reconstructed studying the signal on the segmented anode, while x is reconstructed using the time of the signals.
It has to be noted that TPCs need a uniform field to work properly, which is not an easy feature to achieve, so the setup should be planned accordingly and any non-uniformity must be studied and taken into account. The same goes for drift velocity, which has to be constant and well-known.

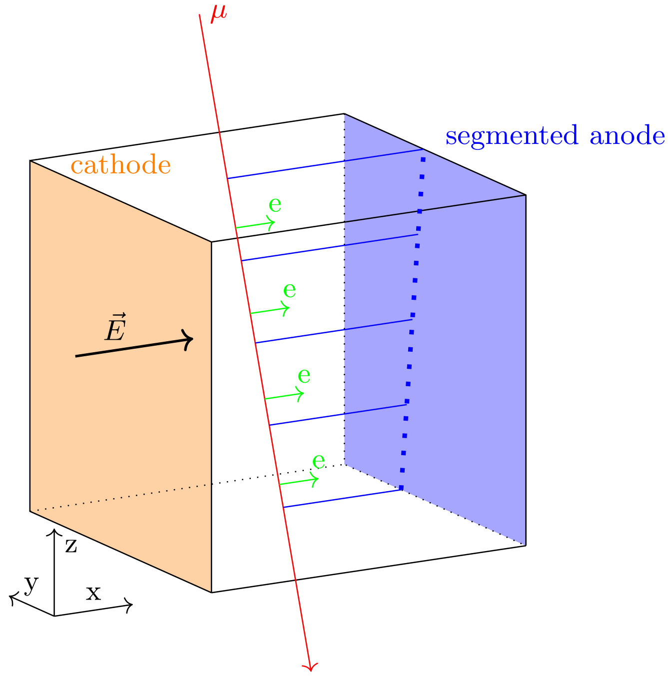


FIG. 1. Scheme of a TPC

II. T2K

The experiment is part of the T2K experimental activity. The T2K experiment [1] in Japan is a famous Long-Baseline collaboration for neutrino physics which seeks to precisely determine the CP violation phase in the neutrino sector. This is achieved by sending a neutrino beam from the JPARC accelerator to the SuperKamiokande observatory. The INFN section of the University of Padua collaborated for the creation of the TPC for the upgrade of the near detector ND280 [2]. The TPC in the ND280 are manly used as trackers for muons produced at high angles due to quasi elastic charged current events. By observing this type of event it is possible to determine the incoming flux of neutrinos from the JPARC accelerator.

The TPC we worked on was one of the prototypes for the field cage in reduced dimensions. Because of the collaboration with members of the T2K experiment, our TPC will resemble the final model, with respect to its shape, gas type and pressure, electric field and other characteristics. This prototype was used to explore the mechanics of the TPC walls and the cathode, and there was no intention of equipping it with a detector and make it fully functional.

III. PREPARATION OF THE APPARATUS

1. Geometry and specifics

The apparatus has a size of 50 cm $\times$ 50 cm $\times$ 47 cm and an estimated weight of over 100 kg. It is divided in two symmetric sections, which we will call forward and backward (F and B), which can be physically separated.

The outer walls are composed of G10 fiberglass, which is a strong dielectric and does not perturb the electric field; inside there are 20 copper strips for every section, which are 6 mm thick and 430 mm long, and the distance between two strips is 4 mm. They were obtained from excavation of a PCB base of 35 μm, and in the process part of the G10 has been removed, so the strip view is as shown in the drawing **2**. In the middle between the two sections, there is the cathode, which is a 14.2 mm thick G10 layer with a PCB of 35 μm on each side, the same as the one used for the strips. The cathode is connected to the G10 outer wall with a pin for each side with a plastic screw.

Between the G10 walls of the two sections, two O-rings are inserted in a slit inside the G10. O-rings are made of rubber and are used as mechanical seals for gas containment. A Teflon sheet is inserted in these slits as well, in order to increase the path length of the charges that may flow from the cathode to the external part of the chamber. Fiberglass screws are used to join the two sections.

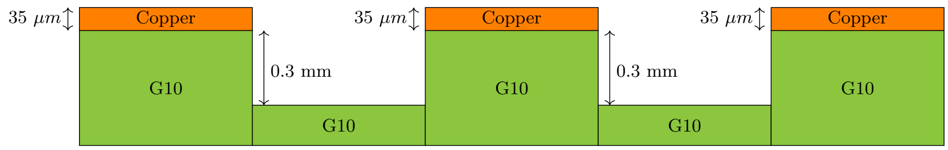
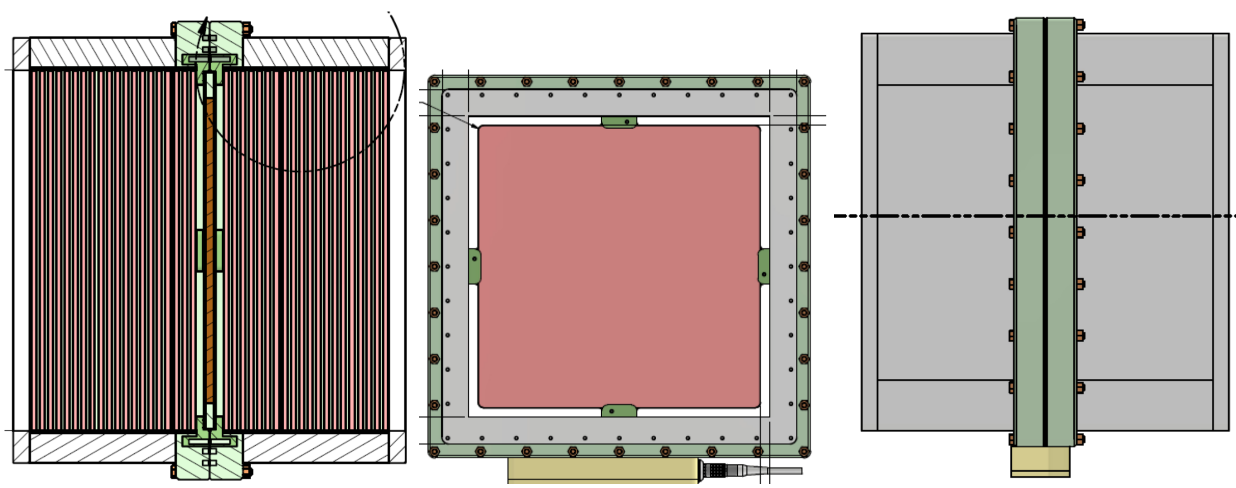


FIG. 2. Scheme of strips in the G10

For each section there is a hole in the G10 between the cathode and the first strip, used for electrical connections; we closed the TPC so that they are both on the same side. The Teflon sheet was perforated in the same area corresponding to the two holes in the G10.
The TPC can be closed by joining the two sections and covering the two open faces with metal frames. NOTA SULLE FRAME DA INSERIRE We provide the technical drawing in figure 3 for a better understanding of the configuration.



(a) Internal section, from the side **(b)** Internal section, from the front **(c)** Exterior of the closed TPC

FIG. 3. Technical drawing of the TPC

2. Coordinates and frame of reference

We defined a frame of reference for the TPC that will be used throughout the report. The zero is positioned in one of the G10 external wall corner that is in the contact edge between the two cages. The specific corner and the direction of the axis is defined by the position of the HV feedthrough. The x direction is perpendicular to the cathode and negative values are on the side where the feedthrough is; the z axis is perpendicular to the ground and the y axis is horizontal. It has to be noted that this system is right-handed, and that the centre of the TPC corresponds to the coordinates (0, 25, 25) cm. The figure 4 shows the reference system. The cathode is marked in orange.

3. Cleaning and soldering procedures

Figure 5 shows the strips at the beginning of our procedure. The strips on the four sides are not electrically connected, so they have to be soldered; moreover, it is necessary to deep clean the TPC, since we do not know how long it has been without proper maintenance and it is covered in copper oxide and has excessive glue on the strips, which will prevent a proper soldering of the components. The glue was used to fix the PCBs to the G10 surface, and the excess had covered the corners and part of the strips, while the oxide was present because the copper parts were exposed to the air humidity and they had been touched bare handed. For this reason, we made sure to wear gloves when touching the copper surfaces.
We started the cleaning by removing the excessive glue from the corners and the strips using files, paying attention

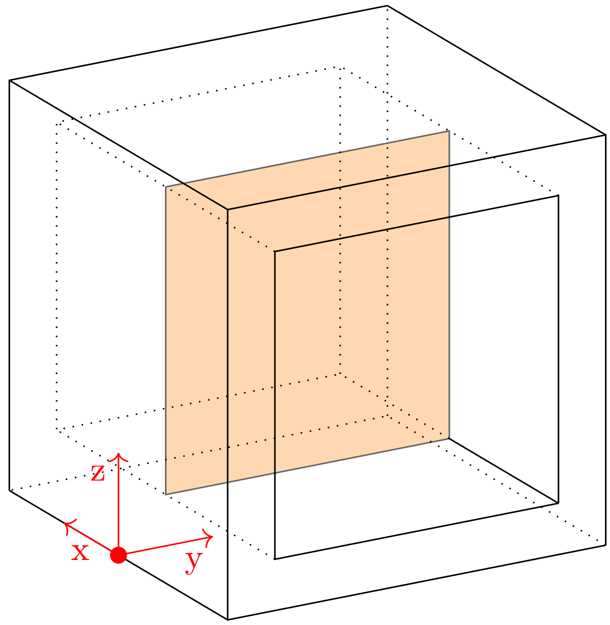


FIG. 4. Reference system drawing

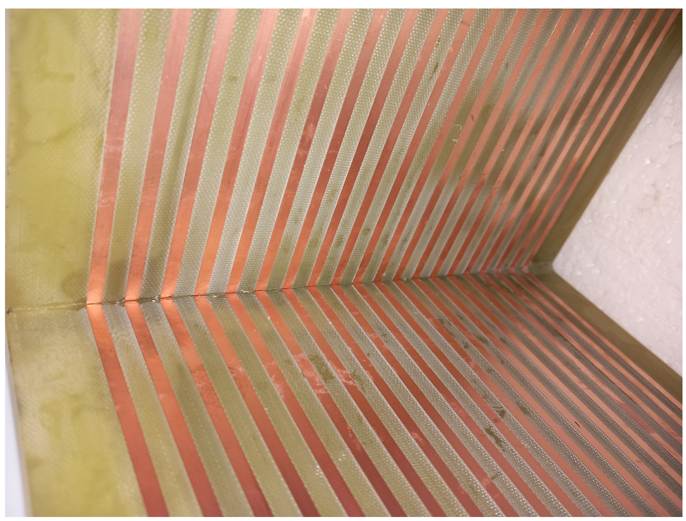


FIG. 5. Picture of the strips before soldering

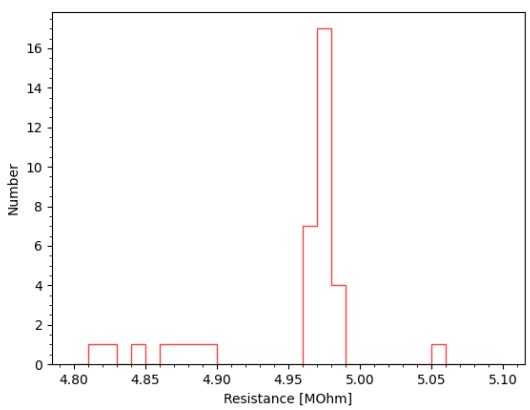
not to damage the strips; then we removed the copper oxide with very thin sandpaper and high purity acetone, in order to prepare the surface to the subsequent soldering. Using the sandpaper produces oxide dust, which may be a problem since it is conductive and may create unwanted contacts between the strips. To prevent this, the dust was thoroughly removed using acetone and blowing it away with compressed air; the procedure was repeated several times, and the TPC was finally cleaned with acetone once again to remove any oil residue on the strips and the TPC. The same was done for the cathode.

We then soldered the four corners of each strip together, in order to have electric contact. The tin layer of the soldering is quite big and curved, since we want to reduce the corona effect in corners. Moreover, the gap between the strips was variable, with a range of some fractions of a millimeter: the round and wide soldering allowed us to connect the strips event in the presence of a wider gap. The resistance between the soldered copper surfaces was measured to check for a good electrical contact and it was always less than 0.2 Ω, the multimeter minimum value.

After this, we soldered 19 resistances between the strips to create the resistive divider for each side. The resistances are of the type "surface mounted device" by Ohmite [3] of nominal value 5 $M\Omega$ with 5% tolerance. The resistances has a length of 5 mm and height of 1 mm; at the short side two electrodes (Nickel-barrier/matte tin) are connected

to the non-inductive serpentine pattern, and this area is the one we soldered to the strips. METTIAMO FOTO RESISTENZE SALDATE The working voltage is maximum 6 kV and they have low temperature and voltage dependency. The resistive bridge for the positive x section is on the right vertical face of the TPC (xz plane), close to the upper right corner; in coordinates, it is along a line perpendicular to the cathode with $y = 468.5mm$ and $z \approx 40cm$. For the negative x section, it is on the same side but close to the lower right corner; in coordinates, it belongs to the line perpendicular to the cathode, with $y = 468.5mm$ and $z \approx 10cm$. The distance of the resistive bridges from the corner is different because of difficulties in soldering.
After soldering, the resistive bridge was cleaned again with acetone, to remove the flux left from the soldering, and all resistances were measured to determine the effective value and the histogram of figure ?? was obtained: all values are within the tolerance.

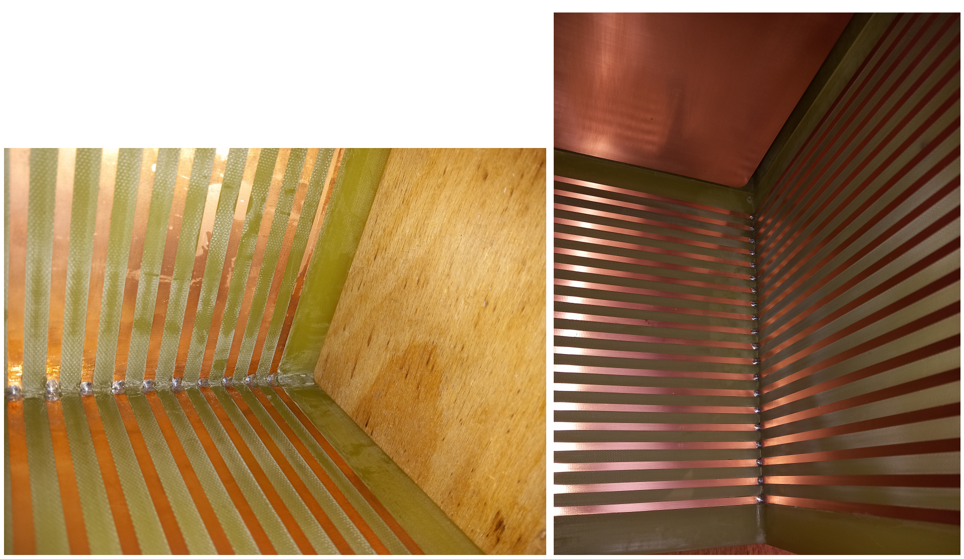
A very deep cleaning of all the strips and the cathode was performed, using sandpaper and acetone. Finally, the



(a) Histogram of the resistance distributions for all 38. The 5% tolerance is $0.25M\Omega$.

(b) Soldered resistances in the TPC.

cleaned cathode was mounted into place and the TPC sections were joined. A visual comparison is provided in figure 7, where the cathode can be seem at the top of the picture.



(a) Before the cleaning **(b)** After the cleaning

FIG. 7. Strips before and after the cleaning and soldering

4. Electrical cabling

The second process of the preparation was the setup of the electrical connection in order to power the device. We used the same electric field as the T2K project, since around this value the drift velocity as a function of the electric field is almost constant; its value is $\approx 270\ V/cm$. Being our chamber $\approx 20\ cm$ long, we need a $5.4\ KV$ potential between the cathode and the last strip.
The electrical connections were done using a feedthrough, which is a component that allows the creation of electrical connections while guaranteeing gas containment. Three feedthrough were used: one for the cathode and two for the last strip. They were created by taking a hollow plastic screw, inserting a copper wire and pouring inside a two phase epoxy resin glue. After solidification we inserted the screw in the threaded hole.
For the high voltage, a support structure attached to one external G10 screw was built. We soldered an HV female connector to the copper wire which was properly insulated with thermoplastics. Then, a support for the female connector was built on the screw, so that the HV could be easily connected.

REFERENCES

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[2] K. Abe et al. T2k nd280 upgrade – technical design report, 2020.
[3] Ohmite Manufacturing Company. Hvc series - high voltage resistors, 2025. Accessed: 2025-04-03.